Quantifying Economic Impacts of Sea Level Rise

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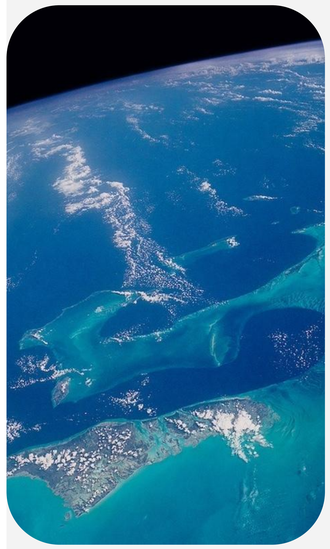
Intro to Sea Level Rise: An Imminent Threat



Process:

1. Humans are emitting **greenhouse gases (GHGs)** into the air, causing **anthropogenic climate change**
2. One factor of climate change is **sea level rise (SLR)**
   - Thermal expansion of oceans
   - Melting of glaciers and ice sheets
3. This will cause threats and damage to coastal settlements and economy
   - Predicted cost of **$1.2-6.9 trillion per year**
   - **Billions of people** worldwide under SLR risk

Facts:

1. Too late to stop sea level rise (but can minimize it!)
   - Long timescale of response
   - Unavoidable for **100's to 1000's of yrs**
2. Increasing at a constant rate of **~4 mm/yr**
3. If the Antarctic ice sheet fully melts, seas will rise **~57 m**

What we did:

1. Predict regional sea level rise by 2100 using BRICK framework
2. Project corresponding economic impacts via pyCIAM model

Data & Methodology

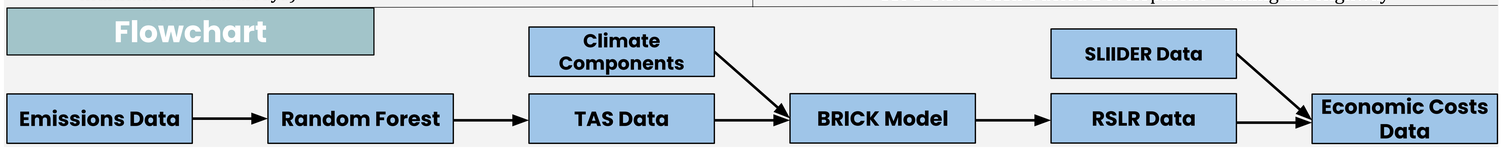
Models/Datasets:

1. **Surface Air Temperature (TAS)** Data
   - Process: **Random Forest** model
   - Inputs: different historical emissions data (**CO_2, CH_4, SO_2, BC**)
   - Outputs: TAS data from **2015-2100**
2. **Regional Sea Level Rise (RSLR)** Data
   - Process: **BRICK** modeling framework.
   - Inputs: **TAS data** and other components that cause SLR
   - Output: RSLR indexed by **latitude**, **longitude**, and **year** from **2015-2100**
     - 1 **NetCDF** file per SSP (multidimensional arrays read by xarray library) and **5,508,000** data points per SSP
3. **Economic Costs** Data
   - Process: **pyCIAM** modeling framework
   - Inputs: **RSLR** data, **SLIIDERS** Dataset (accounts elevation, region, scenario)
   - Output: **Total economic costs** for regions by **2100** in a **Zarr** file (stores multidimensional arrays)

Frameworks:

1. **BRICK Modeling Framework**
   - Breaks down global sea level rise into 4 components
   - Converts global SLR to regional SLR via sea level fingerprints
2. **PyCIAM Modeling Framework**
   - Outputs 6 cost categories which we summed up (capital, life, relocation, protection, inundated land, wetlands)
   - Simulations based on different adaptation scenarios and time periods
3. **Shared Socioeconomic Pathways (SSPs)**
   - Each represent different trajectory humans can take in regards to population growth, economic development, emissions, etc.
   - Models iterated over each SSP we used below
     - SSP1-2.6: "Sustainability - Taking the Green Road"
     - SSP2-4.5: "Middle of the Road" - Current Path
     - SSP3-7.0: "Regional Rivalry - A Rocky Road"
     - SSP5-8.5: "Fossil-Fueled Development - Taking the Highway"

Flowchart



Results & Visualizations

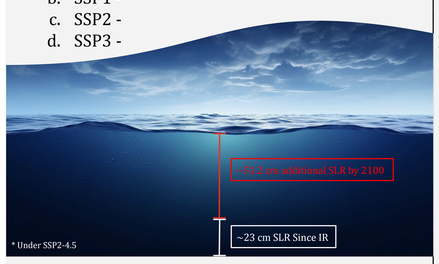
Metrics: TAS, RSLR, and **Costs**

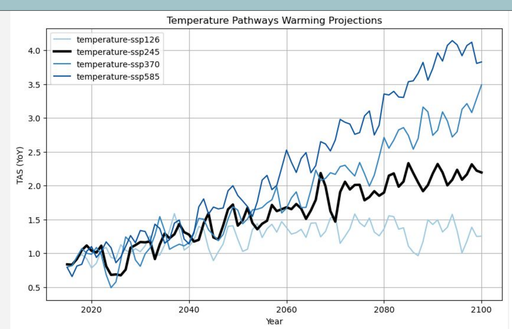
Key Variables

1. **SSPs**
2. **Adaptation** (Ex. "Protect10")
   - Retreat from or protect coast
   - Years of action

Results

1. TAS predicted to increase **2.2°C** from pre-industrial revolution (IR) levels by 2100
2. Global Mean Sea Level Rise
   a. Regions severely impacted
   b. SSP1 -
   c. SSP2 -
   d. SSP3 -
3. Costs
   a. Regions severely impacted: India, SE Asia, Oceania, Eastern SA
   b. SSP1 -
   c. SSP2 -
   d. SSP3 -





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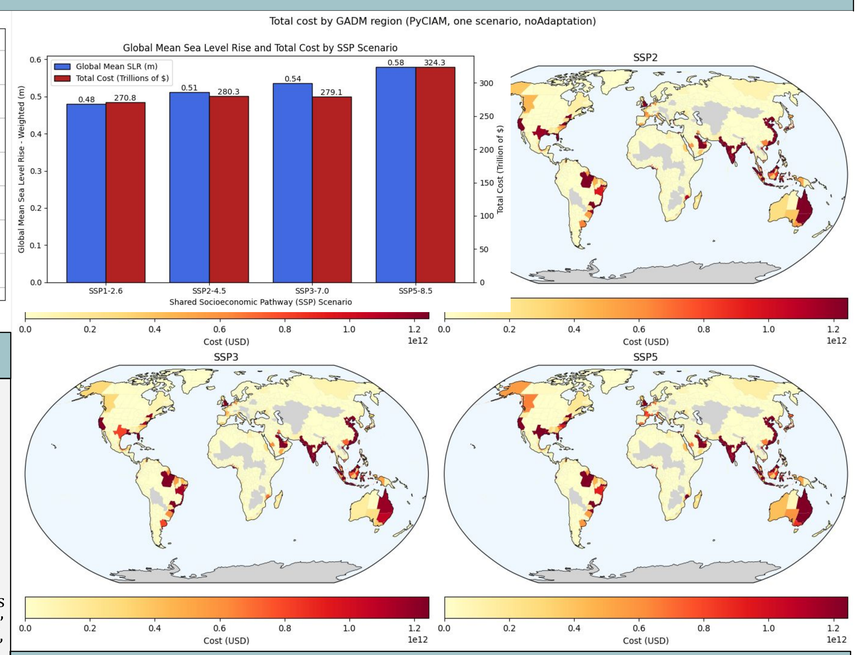
Graphs

1. **TAS** for different SSPs
2. Choropleth maps of **RSLR** and their corresponding **costs**
   a. Current Projections, Delayed Adaptation - SSP2, "Retreat10"
   b. Current Projections, Immediate Adaptation - SSP2, "Protect100"
   c. Increased Emissions, Delayed Adaptation - SSP3, "Protect10"
   d. Difference Between Optimal and Worst-Case Scenarios
      i. Optimal Scenario: SSP1, Adaptation: 'OptimalFixed'
      ii. Worst-Case Scenario: SSP5, Adaptation: 'Retreat10'



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Implications & Future Steps

Implications of Results

- Development of issues in our lifetime
- Catastrophe in centuries to millennia after
  - Sea level projected to rise
  - Major coastal cities and islands **underwater**

Taking Action

1. Reducing Carbon Footprint
   - **Carbon Footprint**: total amount of GHGs emitted by person
2. Electing Policymakers supporting cause against climate change
   - 2015 Paris Climate Agreement:
3. Spreading the Word
   - More people with knowledge about SLR and how to combat it

Conclusion

It's our responsibility to provide a sustainable world for future generations and vulnerable coastal populations. Our mistakes should not be their problems. Let's take the leap to minimize sea level rise before it's too late!